Uniswap v3 Liquidity Provision: Which Range Strategy Earns the Most Against Holding?

An event-level backtest of 11.97 million swaps in the USDC/WETH 0.05% pool, July 2021 to October 2026

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The problem. On Uniswap v3, a liquidity provider (LP) deposits two tokens into a pool and earns a fee on every trade, but only while the price stays inside a range the LP chooses. A narrow range earns more fees per dollar; when the price leaves it, the position ends up entirely in the token that fell and stops earning. There is no standard rule for choosing that range or for when to move it. This note asks which rule would have earned the most in the USDC/WETH 0.05% pool on Ethereum, compared with simply holding the same tokens (HODL), after gas and trading costs.

Key data	
Pool	USDC/WETH 0.05%
Swaps replayed	11,966,957
Period	Jul 2021 to Oct 2026
90-day windows	28 train, 32 test
Rules tested	4 families, 25 settings
Fee error, 30 real positions	0.39% (median)
Best out of sample	Full range, −0.55%
Worst out of sample	Vol. k = 8, −3.22%

Summary

- **No rule beat holding.** Out of sample, full range, rebalance ±90%, fixed ±70% were within noise of holding the same tokens (−0.55% to −1.66% per 90-day window, $|t| < 1.5$). The volatility-scaled rule lost 3.22% ($t = -2.96$).
- **Range width drives the result.** In training, rebalanced ranges of ±10% or narrower lost 4.1% to 33.9% per window, all with t below −2.
- **Rebalancing locks in the loss.** A ±5% range lost 10.5% per window when rebalanced and 2.1% when left alone.
- **The training winner came last.** The volatility rule led training at +0.44% per window and was the only rule clearly below holding out of sample.
- **Costs do not explain the out-of-sample result.** They took 0.01% to 0.12% of a $100k position per window; gas mattered only for small positions in 2021–22.

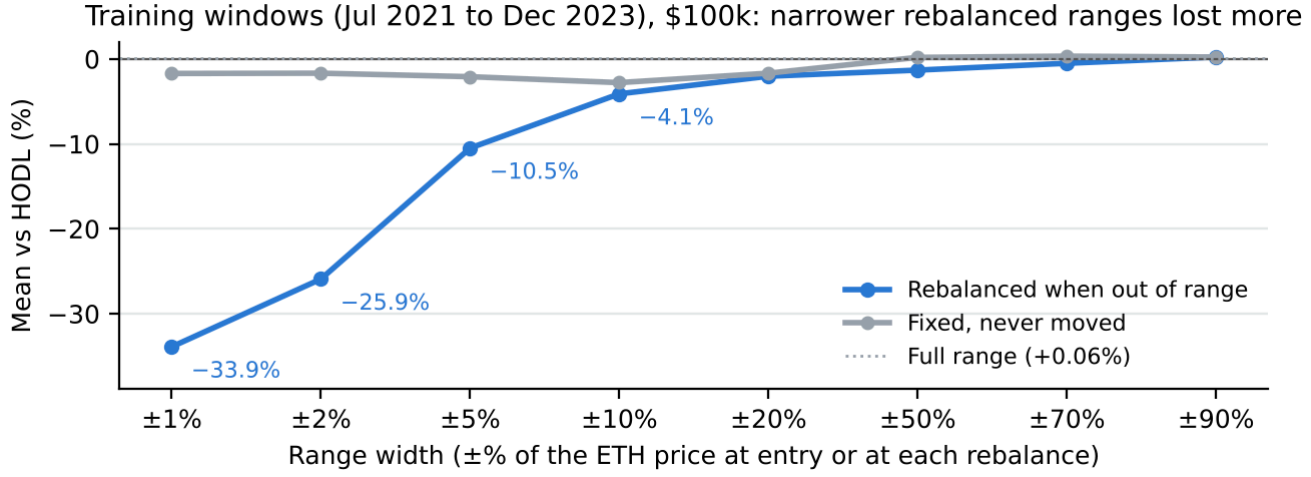


Figure 1. Mean result vs HODL per 90-day window, by range width, 28 training windows each. Source: results/calibration_train.csv.

1. How an LP earns and loses

Fees. Each swap pays 0.05% of its input to the LPs whose range contains the price, in proportion to their liquidity. Concentrating the same capital in a narrower range gives it more liquidity at those prices, so it earns more per dollar while the price stays inside. From 27 December 2025 the protocol keeps a quarter of the fee, so LPs receive 0.0375%.

Impermanent loss. The pool rebalances the LP automatically: as ETH rises it sells ETH, and as ETH falls it buys. Before fees, the position therefore always ends at or below the value of holding the same tokens, and the gap grows with the size of the move (Figure 2). A narrow range loses faster, and once the price leaves it the position is entirely in one token. Milionis et al. (2022) describe the same cost as loss-versus-rebalancing: the LP keeps trading with arbitrageurs at prices the market has already moved past. An LP profits only if fees exceed that loss.

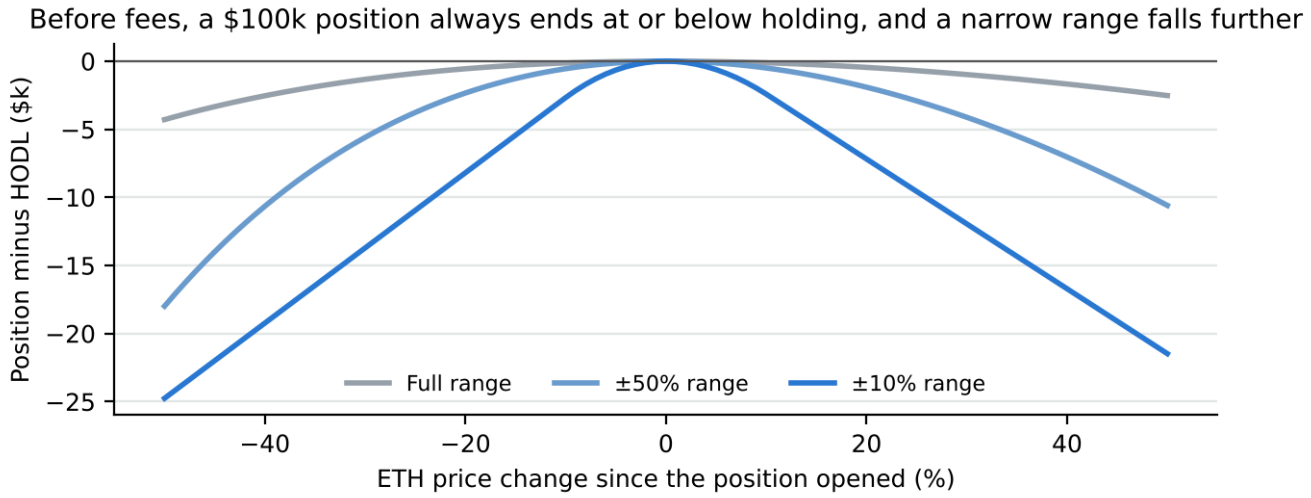


Figure 2. Value of a $100k position minus the value of holding its entry tokens, before fees and costs, by ETH price change. Computed with the backtest's own position formulas.

Rebalancing. When the price leaves a range, an LP can withdraw, swap back to the right token mix and open a new range around the price. That restarts fee income, but it turns the gap into a realised loss and pays gas, a 0.05% swap fee and price impact each time. The question is which combination of width and rebalancing leaves fees ahead of these losses.

Why July 2021. The pool opened on 5 May 2021. In late May a trade of a few thousand dollars could move its price by 1%, and that depth reached about $1M only in mid-June, so a $100k position would have been a large share of the pool. From July 2021 no gap between consecutive swaps exceeds 149 blocks and no day has fewer than a quarter of the median 5,592 swaps per day.

2. Data, model and validation

A replay engine inserts a hypothetical position into every real swap, in chain order. On each swap that ends inside its range, the position earns the input amount × 0.05% × (1 − protocol share) × its share of the active liquidity. About once an hour the engine values the position and, if the rule says so, moves it. Each window opens a new position on the first day of the month and holds it for 90 days; its result is (position + fees − HODL value) / initial capital, where HODL keeps the position's entry token mix and pays no gas.

Rule	Range	When the price leaves it	Settings tested
Full range	All prices (as Uniswap v2)	Never leaves	—
Fixed	±x% of the entry ETH price	Nothing	x = 1–90%
Rebalance	±x% of the current price	New range around the price	x = 1–90%
Volatility	±k daily std. dev. (last 7 days)	New range, new width	k = 0.5–12

Element	Choice	Source
Swap events	11,966,957 swaps, May 2021 to Oct 2026	Pool 0x88e6…5640, Etherscan API V2
Gas price	46,488 samples, one per 300 blocks (≈1 h)	Median price paid in the sampled block
Gas used	mint 418,892 / burn 243,676 / swap 192,352	Medians of 430 real transactions
Protocol fee	LPs keep 0.0375% instead of 0.05% from block 24,106,471	SetFeeProtocol event (27 Dec 2025), slot0
Price impact	Swap size / pool depth	Median active liquidity of the last 500 swaps
Checks	Every 300 blocks, at the end of a block	Ignores sandwiches inside one block
Windows	90 days, opened monthly; $10k, $100k, $1M	Train Jul 2021–Dec 2023, test Jan 2024–Oct 2026
Selection / significance	Highest mean vs HODL in training	Newey-West t-statistics, 2 lags

Validation. The fee calculation was checked against positions that existed on-chain, and the whole engine against an independent implementation:

Check	Result
Fees vs 30 real positions, 2021–26 (Collect − DecreaseLiquidity)	Median error 0.39%; total +0.32%; worst 5.22%
Positions after the protocol fee switch	Errors of 0.08% or less
Position amounts vs Uniswap's own test cases; impermanent loss vs the v2 formula	Exact
Independent re-implementation in ETH-price space	Same result to six decimals in four windows
Consistency of all saved windows	No violations in 1,500 windows
Review by a separate reviewer	No coding errors; two readings and the selection rule revised

The largest fee error (5.22%) is a position only 40 ticks wide, where swaps crossing the range boundary are approximated. Rebalancing costs are modelled from real gas and pool depth but were not checked against a real rebalancing LP.

3. Results

Training: width decides. Figure 1 shows the pattern. Rebalanced ranges lose more the narrower they are: ±1% rebalanced 234 times per window and lost 33.9%, of which 26.5% was gas, swap fees and price impact. Fixed ranges that never move lose less but vary widely (worst window −31.7%), so none of their means is distinguishable from zero. The volatility rule follows the same pattern by k, from −26.2% at k = 0.5 to +0.4% at k = 8. The best setting of each family was the widest or close to it.

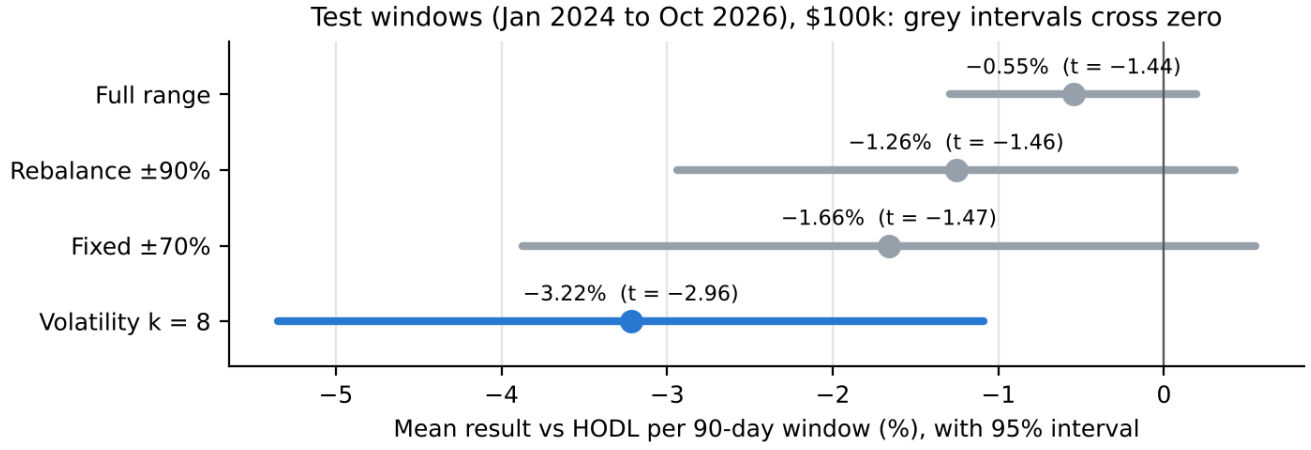


Figure 3. Out-of-sample mean result vs HODL per window with mean ± 1.96 Newey-West standard errors, 32 test windows each, parameters chosen in training. Source: results/calibration_test.csv.

Out of sample: a tie at best. Every mean is negative. Three rules are statistically indistinguishable from holding; only the volatility rule, first in training, is clearly below. Full range beat holding in 18 of 32 windows; one window opened on 2025-05-01 lost 8.6% and accounts for much of its mean (without it, −0.29%). For a wide range, fees roughly offset impermanent loss on average; an LP should expect no edge over holding, not a reliable loss or gain.

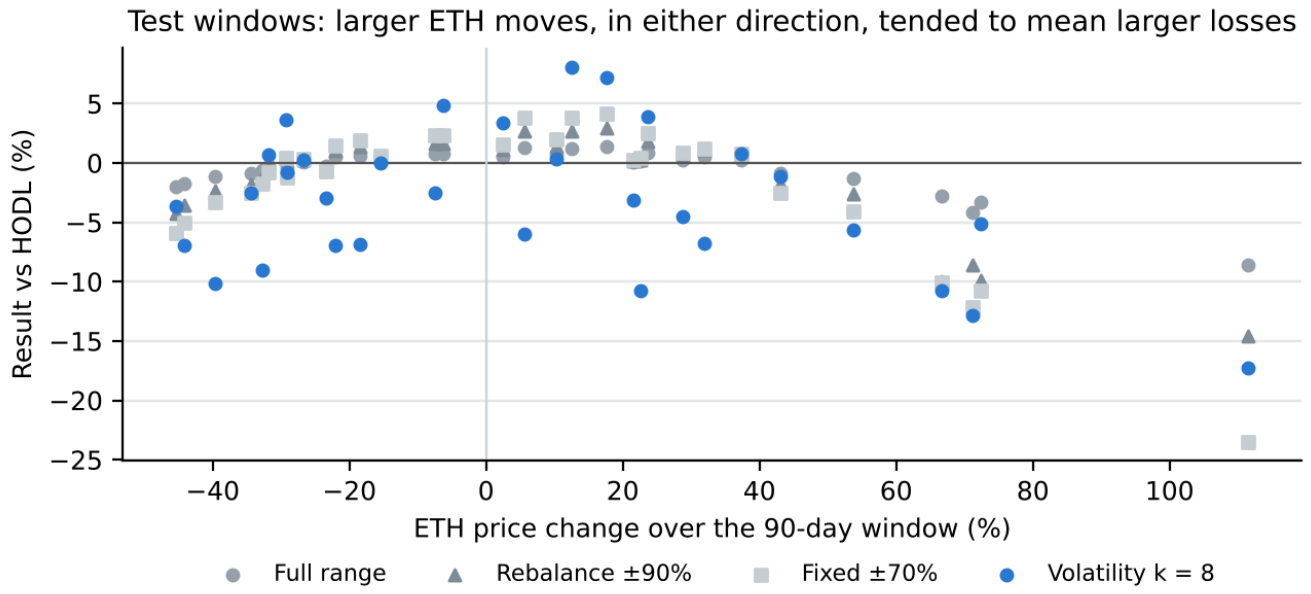


Figure 4. Result vs HODL of each test window against the ETH price change over that window (twice the HODL 50/50 return). Source: results/calibration_test_windows.csv.

What drives the loss. Figure 4 is Figure 2 with real fees added: windows with a small ETH move end near or above holding, and large moves in either direction end below it. Rules that hold a narrower range, such as the volatility rule, fall further for the same move.

By regime. Mean result vs HODL per window, $100k (t shown where |t| ≥ 2; regimes before 2024 include training data):

Regime	Windows	Full range	Rebalance ±90%	Fixed ±70%	Volatility k = 8
2021 H2 (rally)	4	−0.05%	+0.21%	+0.35%	−1.35%
2022 (drawdown)	10	+0.23%	+0.56%	+0.75%	+0.06%
2023–24	22	−0.04%	−0.04%	−0.09%	−1.19%
2025–26	20	−0.96%	−2.20%	−2.89%	−3.09% (t = −2.02)

By position size. Gas is a fixed amount per transaction, so it weighs on small positions when it is expensive. The average gas price paid fell from about 79 gwei in 2021 and 44 gwei in 2022 to 0.48 gwei in 2026. Full range:

Regime	Costs, $10k	Costs, $100k	Result vs HODL, $10k
2021 H2 (rally)	2.69%	0.27%	−2.68% (t = −3.88)
2022 (drawdown)	0.86%	0.09%	−0.48%
2023–24	0.32%	0.03%	−0.35%
2025–26	0.02%	0.00%	−0.98%

At $1M results are close to $100k, but less reliable in 2025–26: pool liquidity fell about 7-fold from 2023 to 2025, so a $1M position becomes a large share of the pool, and fees from swaps where it would hold more than 5% of the liquidity are uncertain (volatility rule: −3.6% with them, −4.8% without).

4. Risks and limitations

What could change the result. (1) **Fee share** uses the liquidity after each swap, so a swap that crosses a liquidity gap credits the position with too much; this favours the LP and does not weaken the conclusion, but makes $1M results in 2025–26 less reliable. (2) **Rebalancing is optimistic**: it executes at the triggering end-of-block price with no delay and no MEV. (3) **Small effective sample**: overlapping windows give roughly 11 independent observations out of sample and 9 in training; 2021 H2 has 4 windows. (4) Fees are not reinvested. (5) Before the September 2022 Merge, 300 blocks were about 68 minutes, so hourly checks were slightly less frequent.

Consistency with the literature. Loesch et al. (2021) found that LPs in 17 Uniswap v3 pools would have been $60.8m better off holding. This backtest reaches the same direction with rules an LP could follow, over five years and out of sample.

Next steps. A hedged LP, shorting ETH perpetual futures against the position's WETH, removes the direction of the move but not the convexity in Figure 2; it should be compared with holding USDC, net of funding. Other rule families (asymmetric ranges, a wide base plus a narrow range, scheduled rebalancing) and other pools, such as the 0.3% tier or layer 2 networks, are untested.

Sources. Pool events, gas prices and receipts: Ethereum mainnet via Etherscan API V2. Loesch, Hindman, Richardson and Welch (2021), "Impermanent Loss in Uniswap v3", arXiv 2111.09192. Milionis, Moallemi, Roughgarden and Zhang (2022), "Automated Market Making and Loss-Versus-Rebalancing", arXiv 2208.06046. Uniswap, "Protocol Fee Configuration", docs.uniswap.org. Full code, tests and results: github.com/ChristopheAbelem/uniswap-lp-backtest.